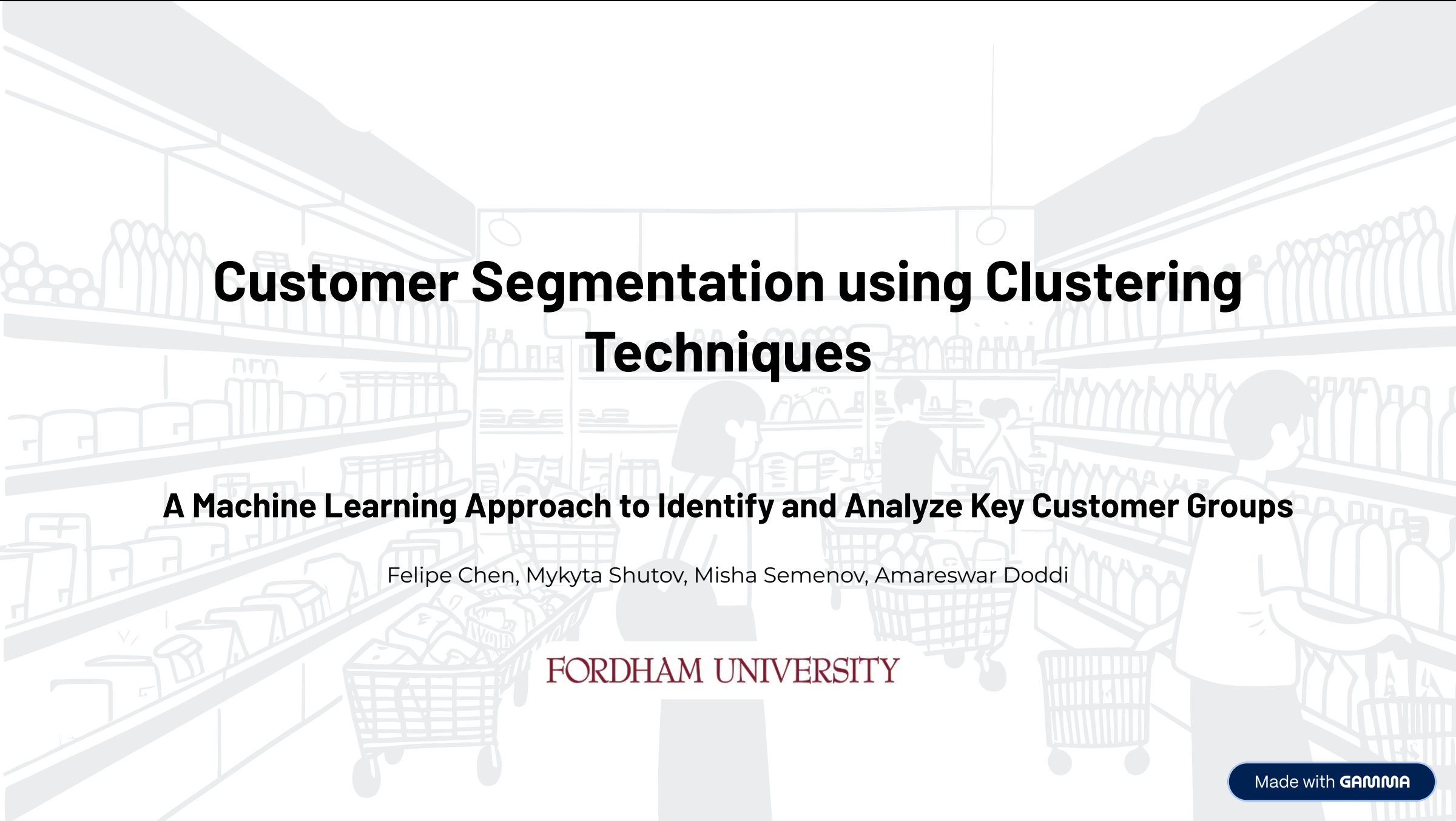




Customer Segmentation using Clustering Techniques

A Machine Learning Approach to Identify and Analyze Key Customer Groups

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Introduction & Objective

Objective : Explore customer purchasing behavior and identify key characteristics of the best buyers.

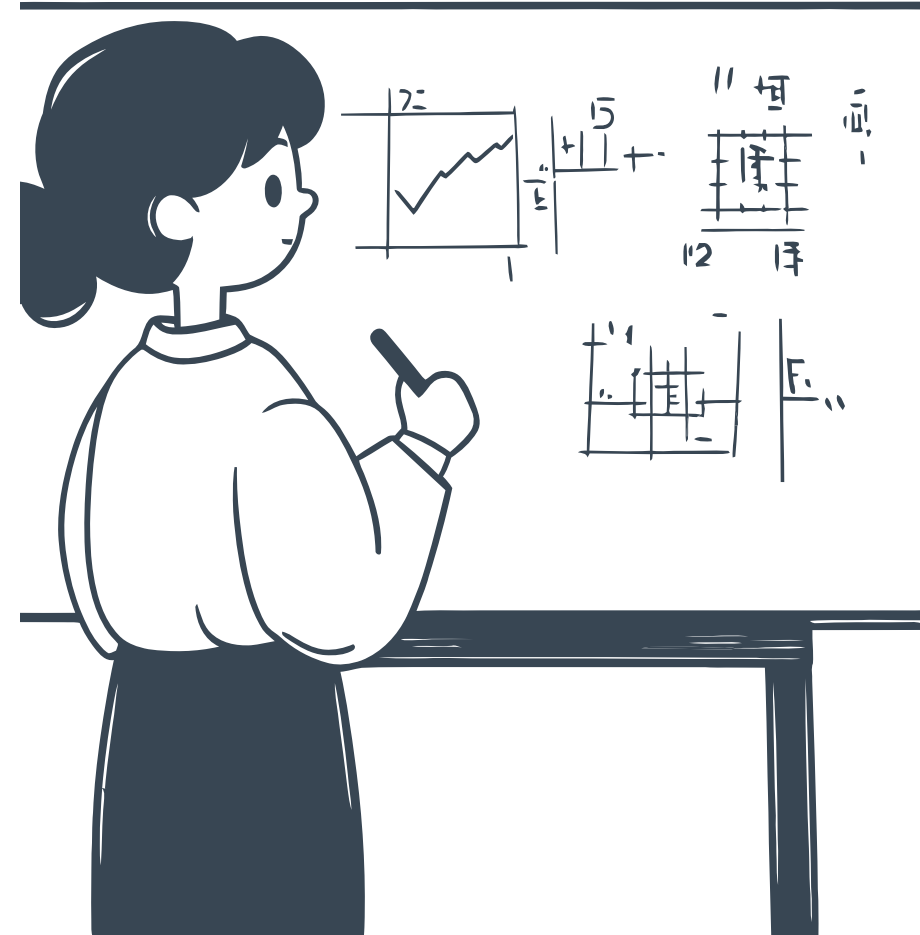
Importance : Understanding which customers contribute most to sales can help businesses develop targeted marketing strategies.

Method : Applied Unsupervised learning (K-Means and Hierarchical Clustering)

Problem-Solving Statement

Businesses struggle to identify which customers generate the most value and how to target them effectively.

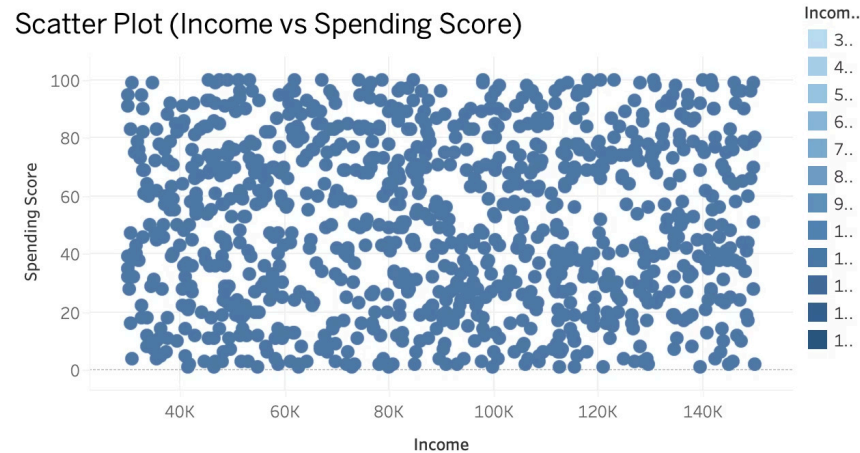
- The goal is to use unsupervised ML (clustering) to group customers based on their spending patterns, income, and preferences.
- These insights can help identify high-value segments and guide marketing and retention strategies.



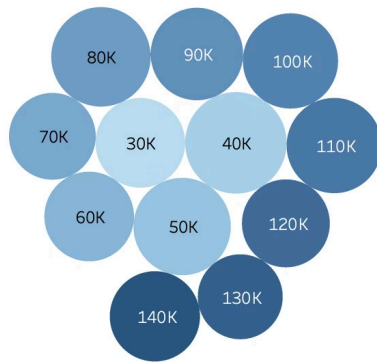
Dataset Description

Feature	Type	Description
id	Integer	Unique customer identifier
age	Integer	Customer's age
gender	Categorical	Gender of the customer (Male, Female, Other)
income	Integer	Annual income of the customer
spending_score	Integer	Score representing customer's spending habits (1–100)
membership_years	Integer	Years of membership with the company
purchase_frequency	Integer	Number of purchases per year
preferred_category	Categorical	Product category most frequently purchased
last_purchase_amount	Float	Amount spent in the last purchase

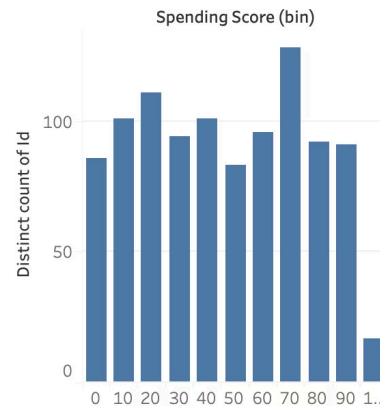
Exploratory Data Analysis (Before Clustering)



Bubble Chart (Income Distribution by Bins)



Histogram of Spending Score



- **Scatter Plot (Income vs Spending Score) :**
Spending behavior varies widely across all income levels, indicating no simple linear relationship between income and spending score.
- **Bubble Chart (Income Distribution by Bins) :**
Most customers fall into mid-to-high income ranges (50K–110K), with fewer customers at extreme low- or high-income levels.
- **Histogram of Spending Score (Binned) :**
Spending scores are spread evenly across the 0–100 scale, showing multiple customer behavior types.

Methodology (*Project Workflow*)

1

Data Collection

Downloaded dataset from Kaggle.

2

Data Preprocessing

Encoded categorical data,
normalized numeric fields.

3

Feature Selection

Chose *Income* and *Spending Score*
for clustering.

4

Model Building

Applied K-Means and Hierarchical
clustering.

5

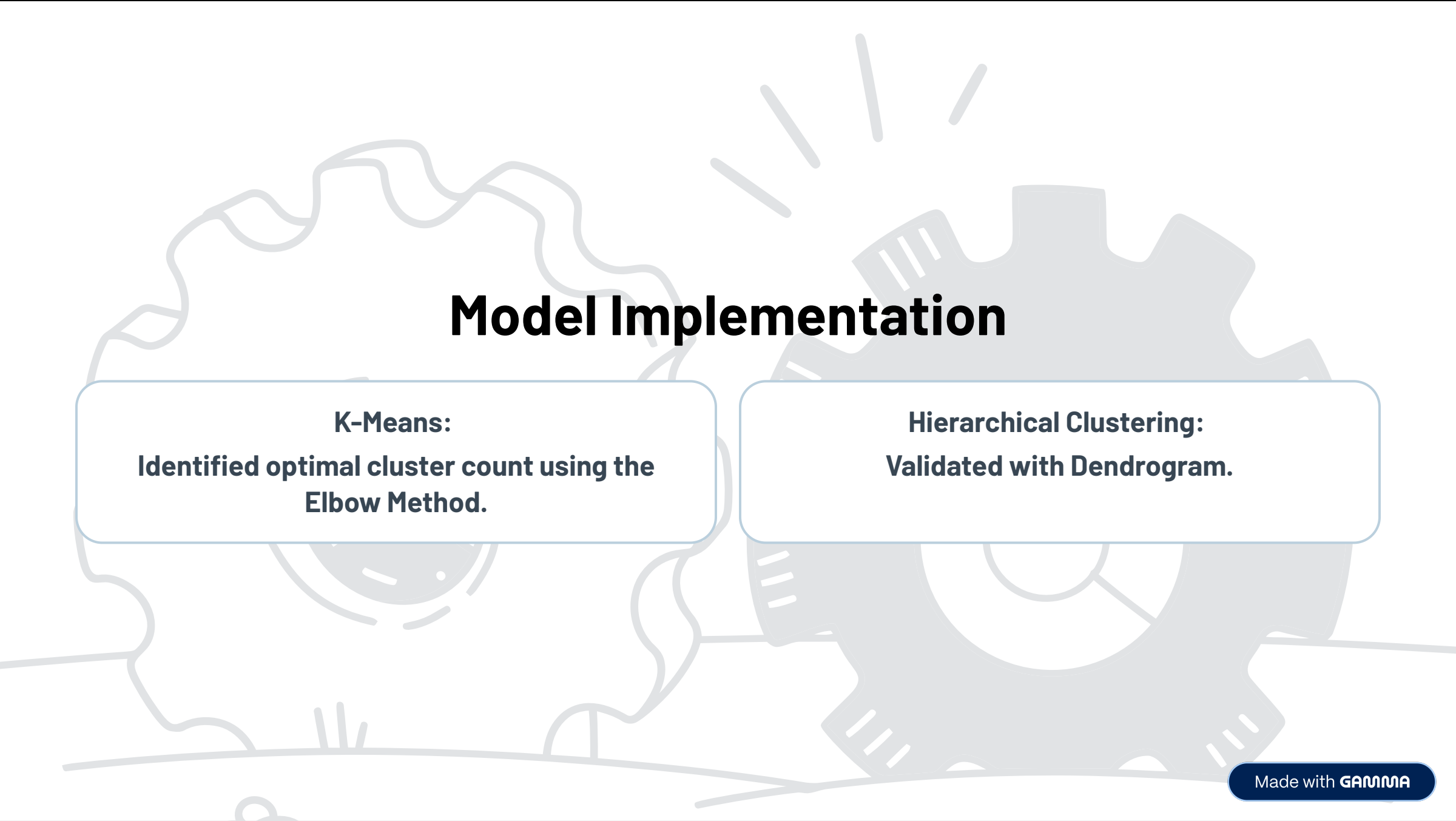
Evaluation

Used Elbow Method, Inertia, and
Silhouette Coefficient.

6

Visualization

Used Tableau for Charts



Model Implementation

K-Means:

Identified optimal cluster count using the Elbow Method.

Hierarchical Clustering:

Validated with Dendrogram.

How K-Means Clustering Works:

01

K-Means groups customers into **four** segments based on **income** and **spending behavior**.

02

Each cluster represents a unique **customer persona** with distinct purchasing patterns.

03

Cluster centroids highlight the **average** profile of each group, improving interpretation.

04

K = 4 was selected because it clearly separates high-value, medium-value, and low-value customers.

05

This **segmentation** supports targeted marketing, cross-selling, retention, and customer value analysis.

06

Clustering reveals **hidden patterns** in customer data that are not visible in simple EDA charts.

How K-Means Clustering Works:

01

```
# Clusters seems to be the best (k=4)
```

```
kmeans = KMeans(n_clusters=4, init='k-means++', max_iter=500, random_state=42)
y = kmeans.fit_predict(X_scaled)
labels_kmeans = kmeans.labels_
labels_kmeans
```

```
# Silhouette score is reasonable
```

```
score = silhouette_score(X_scaled, labels_kmeans)
print(score)
```

0.4189735063734962

Visualization : K-Means Clustering

Cluster 1: High spending score - low income.

Cluster 2: High spending score - high income.

Cluster 3: Low spending score - low income.

Cluster 4: Low spending score - high income.

```
# Getting numerical summary of clusters
df_scaled.astype(float)
df_scaled.groupby('Clusters').mean().round(2)
```

	age	income	spending_score	membership_years
Clusters				
0	43.32	-0.93	-0.92	5.45
1	43.59	0.82	-0.83	5.42
2	44.04	-0.79	0.83	5.61
3	44.19	0.94	0.93	5.36

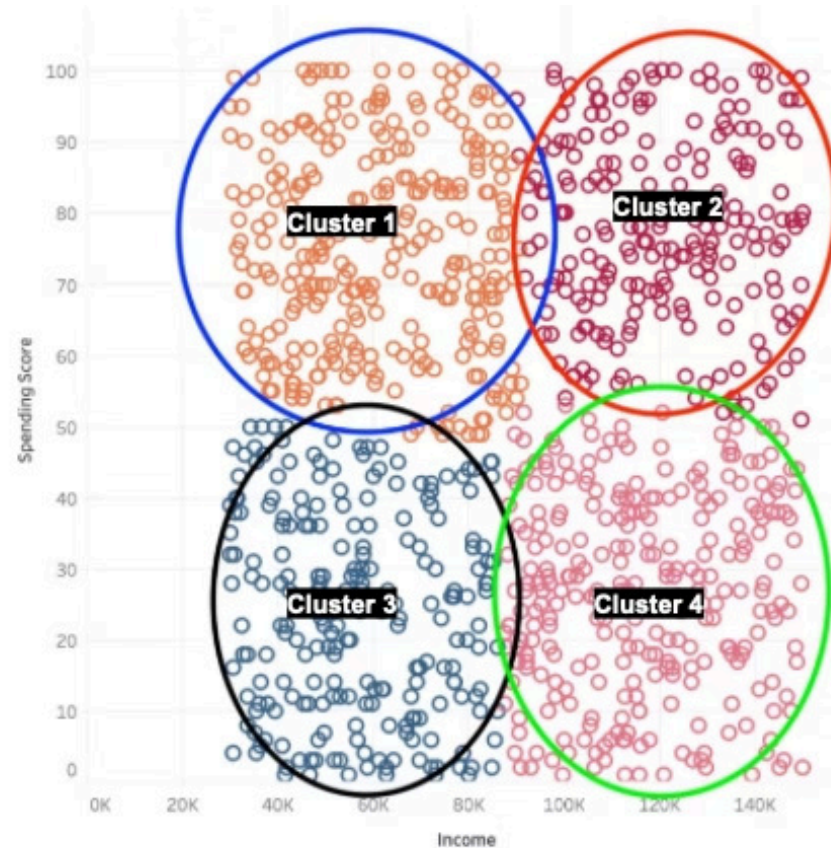


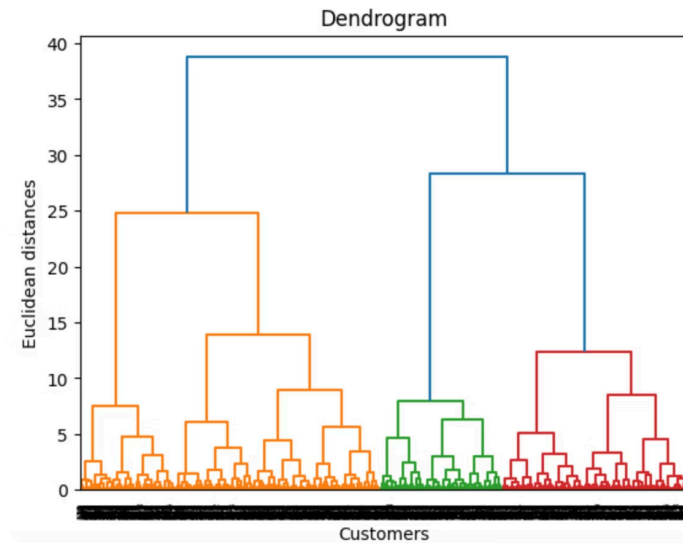
Chart 1: K-Means – Income vs Spending Score

How Hierarchical Clustering Works:

01

```
import scipy.cluster.hierarchy as sch

dendrogram = sch.dendrogram(sch.linkage(X_scaled, method = 'ward'))
plt.title('Dendrogram')
plt.xlabel('Customers')
plt.ylabel('Euclidean distances')
plt.show()
```



```
from sklearn.cluster import AgglomerativeClustering
```

```
hc = AgglomerativeClustering(n_clusters = 5, metric = 'euclidean', linkage = 'ward')
y_hc = hc.fit_predict(X_scaled)
```

Silhouette Score is lower compared to K-means

```
print('Silhouette Score HC:', silhouette_score(X_scaled, y_hc))
```

Silhouette Score HC: 0.37411356233203535

Visualization : Hierarchical Clustering (HC)

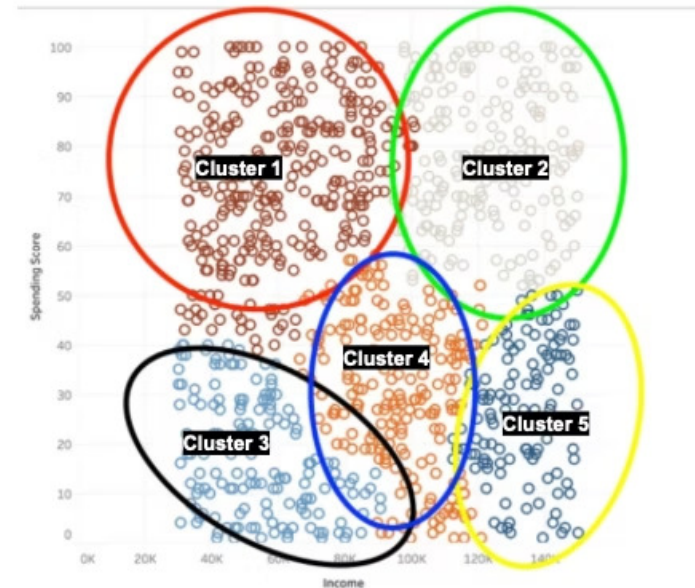
Cluster 1: High spending score - low income.

Cluster 2: High spending score - high income.

Cluster 3: Low spending score - low income.

Cluster 4: Low spending score - middle income.

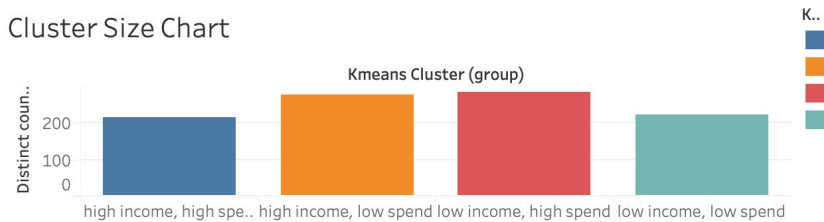
Cluster 5: Low spending score - high income.



	age	income	spending_score	membership_years
HC_Cluster				
0	43.68	-0.82	0.80	5.56
1	43.06	0.22	-0.69	5.39
2	44.97	0.99	0.92	5.38
3	43.56	-1.01	-1.15	5.78
4	43.61	1.32	-0.82	5.13

K-Means Cluster Profiles & Key Segment Insights

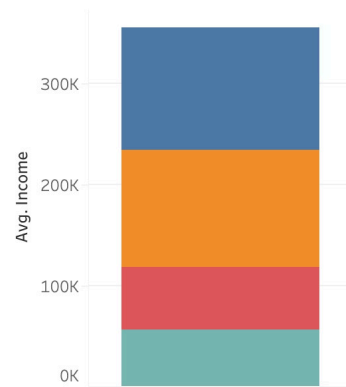
Cluster Size Chart



Average Age



Average Annual Income



Average Spending Behavior Score



- **Cluster Size Chart** : The largest group is Low Income – High Spend, showing strong spending despite lower income levels.
- **Average Age** : Average age is similar across all clusters (early 40s), meaning age is not a major segmentation factor.
- **Average Annual Income** : High Income – High Spend has the highest income, aligning with premium customer potential.
- **Average Spending Behavior Score** : High Income – Low Spend customers represent an opportunity for targeted promotions to increase spending.

Insight from clusters

The data clusters clearly show that **Spending Score** is the most influential and actionable factor in customer segmentation.

It creates two distinct and consistent groups - high spenders and low spenders - more reliably than income does.

From a business perspective, the strategic priority should be to strengthen purchasing behavior within the large low-spending segment. Building a strong brand presence and fostering customer loyalty would be effective approaches to achieving this goal as well as keeping high spending customers.

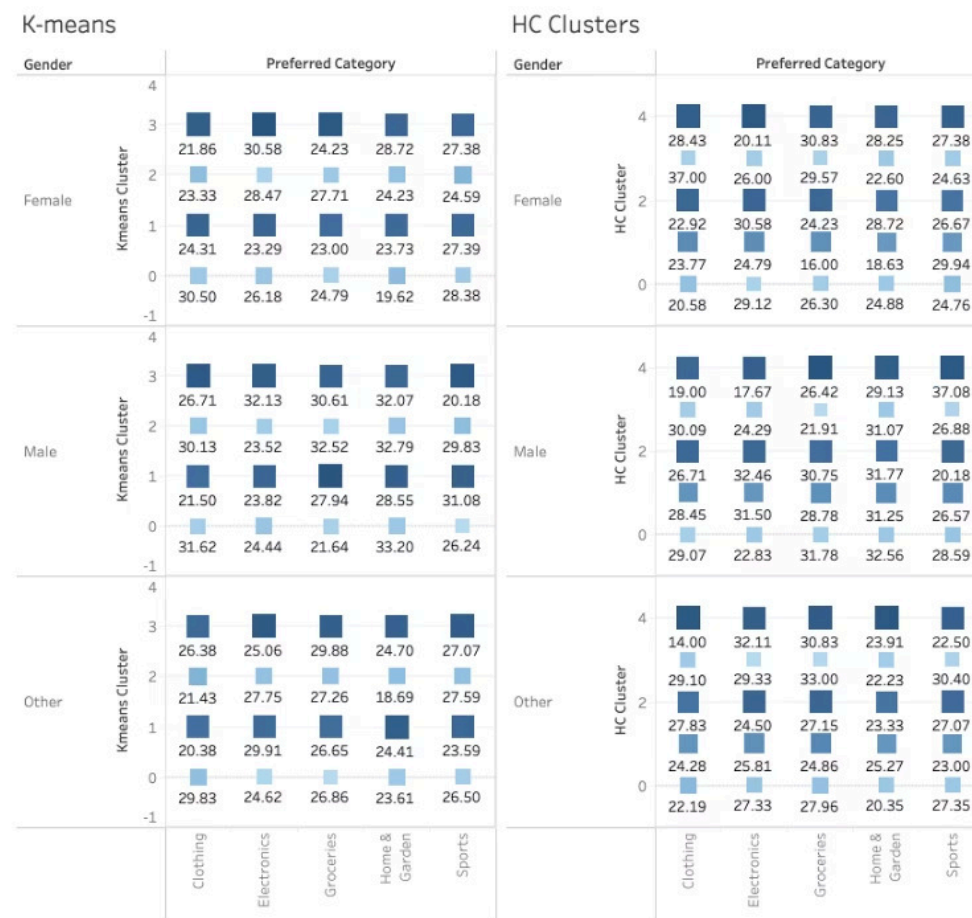


Clustering Visualization Insights

This visualization allows us to examine how each cluster behaves in terms of purchasing preferences. By breaking the clusters down by gender and preferred product category, we can see differences in buying patterns even among customers with similar income and spending profiles.

Across K-means and hierarchical clusters, females, males, and customers identifying as "other" show distinct category preferences, and their purchase frequencies vary noticeably within the same cluster level.

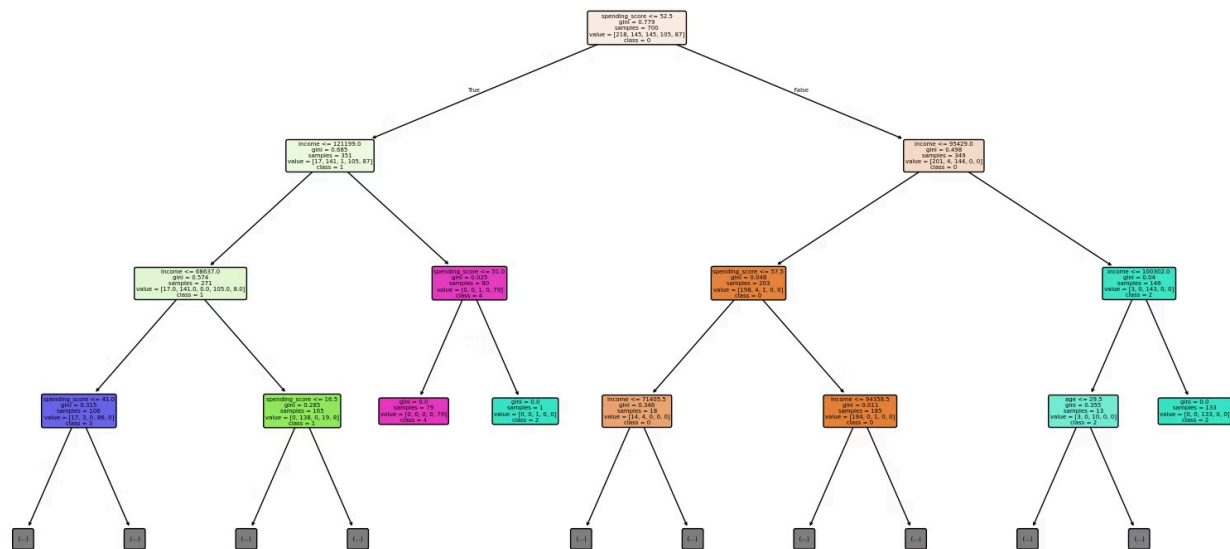
This suggests that demographic factors, especially gender, continue to influence buying behavior independent of income-based segmentation.



Decision Tree Model Insights

A Decision Tree splits the data based on the most important features to predict whether a customer is a **high spender or low spender**.

The tree identifies **Income**, **Spending Score**, and **Age** as key predictors.



Customer Segments Overview



Devoted Fans

High engagement despite lower income



Power Customers

Premium buyers with high value



Emerging Relationships

Early-stage customers with growth potential



Opportunity Accounts

Untapped high-income prospects

Devoted Fans

High Spend / Low Income

Profile

- **Characteristics:** High spending score despite lower income levels
- **Behavior:** Loyal customers who prioritize value and quality over price
- **Size:** Largest customer segment in our data



Marketing Actions



Loyalty Programs

Reward frequent purchases with points, discounts, and exclusive access



Value-Based Messaging

Emphasize quality, durability, and long-term value in communications



Community Building

Create brand communities and user-generated content campaigns



Flexible Payment Options

Offer installment plans and seasonal promotions to support purchasing power

Power Customers

High Spend / High Income

Profile

- **Characteristics:** High income with high spending scores - premium customer segment
- **Behavior:** Quality-focused buyers willing to pay premium prices for superior products
- **Value:** Highest revenue per customer with strong profit margins



Premium Strategies



VIP Treatment

Exclusive access to new products, private sales, and premium customer service



Premium Product Lines

Curate high-end collections and luxury offerings specifically for this segment



Personalized Experiences

One-on-one consultations, custom solutions, and white-glove service



Strategic Partnerships

Collaborate with luxury brands and offer exclusive co-branded experiences

Emerging Relationships

Low Spend / Low Income

Profile

- **Characteristics:** Lower income customers with currently low spending patterns
- **Behavior:** Price-sensitive buyers who need encouragement to increase purchase frequency
- **Potential:** Early-stage relationships with significant growth opportunity



Cost-Efficient Strategies



Entry-Level Products

Offer affordable starter products to build initial purchasing habits



Educational Content

Provide value through tutorials, tips, and product education to build trust



Gradual Engagement

Use email marketing and social media to maintain regular, low-cost touchpoints



Referral Incentives

Encourage word-of-mouth growth through friend referral programs and social sharing

Opportunity Accounts

Low Spend / High Income

Profile

- **Characteristics:** High income customers with surprisingly low spending scores
- **Behavior:** Have purchasing power but aren't currently engaged with our brand
- **Opportunity:** Significant untapped revenue potential from affluent but inactive customers



Reactivation Strategies



Targeted Promotions

Create compelling offers specifically designed to re-engage dormant high-value prospects



Premium Positioning

Showcase high-quality products and services that match their income level and lifestyle



Competitive Analysis

Research what competitors are offering and develop superior value propositions



Direct Outreach

Personal sales calls and account management to understand barriers and rebuild relationships

Strategic Takeaways

Our clustering analysis reveals four distinct customer segments that require tailored marketing approaches.

- Devoted Fans represent our most loyal base and should receive loyalty-focused retention strategies.
- Power Customers demand premium experiences and represent our highest-value opportunities for revenue growth.
- Emerging Relationships need cost-effective nurturing to develop their purchasing potential over time.
- Opportunity Accounts present the greatest untapped revenue potential, requiring targeted reactivation campaigns to convert their high income into high spending.

By implementing segment-specific strategies, we can optimize marketing spend, improve customer lifetime value, and drive sustainable revenue growth across all customer groups.



Projected Revenue per Customer Segment

Our analysis projects the total annual revenue at **\$928,000**, broken down by segment as follows:



Power Customers

\$396,000

Our highest value segment, contributing the most to current revenue.



Devoted Fans

\$350,000

Strong contribution through consistent, high-volume purchases, forming a loyal base.



Opportunity Accounts

\$140,000

Currently underperforming relative to their high income, indicating significant untapped potential.



Emerging Relationships

\$42,000

The smallest revenue share, but with strong growth potential through nurturing.

This breakdown highlights that **Power Customers** are our most valuable segment today, while **Opportunity Accounts** represent a significant undermonetized potential given their high income.

Projected Numbers from Power Customer Strategy

Focusing on the [Power Customers](#) segment can yield significant revenue growth through targeted strategies, converting their high income into consistent high spending.

15%

Spending Increase

Assumed annual spending increase

\$396K

Baseline Revenue

Annual revenue from Power Customers

\$455.4K

New Revenue

Projected revenue after uplift

+\$59.4K

Incremental Gain

Additional annual revenue generated

Strategic Levers

VIP Programs

Exclusive access and benefits to foster loyalty.

Early Access

First look at new products and limited editions.

Premium Bundles

Curated packages of high-value products.

Concierge Support

Personalized, high-touch customer service.

Projected Uplift from Opportunity Accounts Strategy

Focusing on the [Opportunity Accounts](#) segment, which comprises high-income customers with low current spending, presents a significant chance for growth. These customers don't need loyalty — they need activation.

30%

Spending Increase

Assumed annual spending increase to tap into financial capacity

\$140K

Baseline Revenue

Current annual revenue from Opportunity Accounts

\$182K

New Revenue

Projected revenue after activating this segment

+\$42K

Incremental Gain

Additional annual revenue generated by strategic focus

Reactivation Strategies for Growth

Targeted Promotions

Craft compelling, personalized offers to re-engage dormant high-value prospects and encourage initial purchases.

Personalized Outreach

Utilize direct communication channels and account management to understand specific barriers and rebuild relationships.

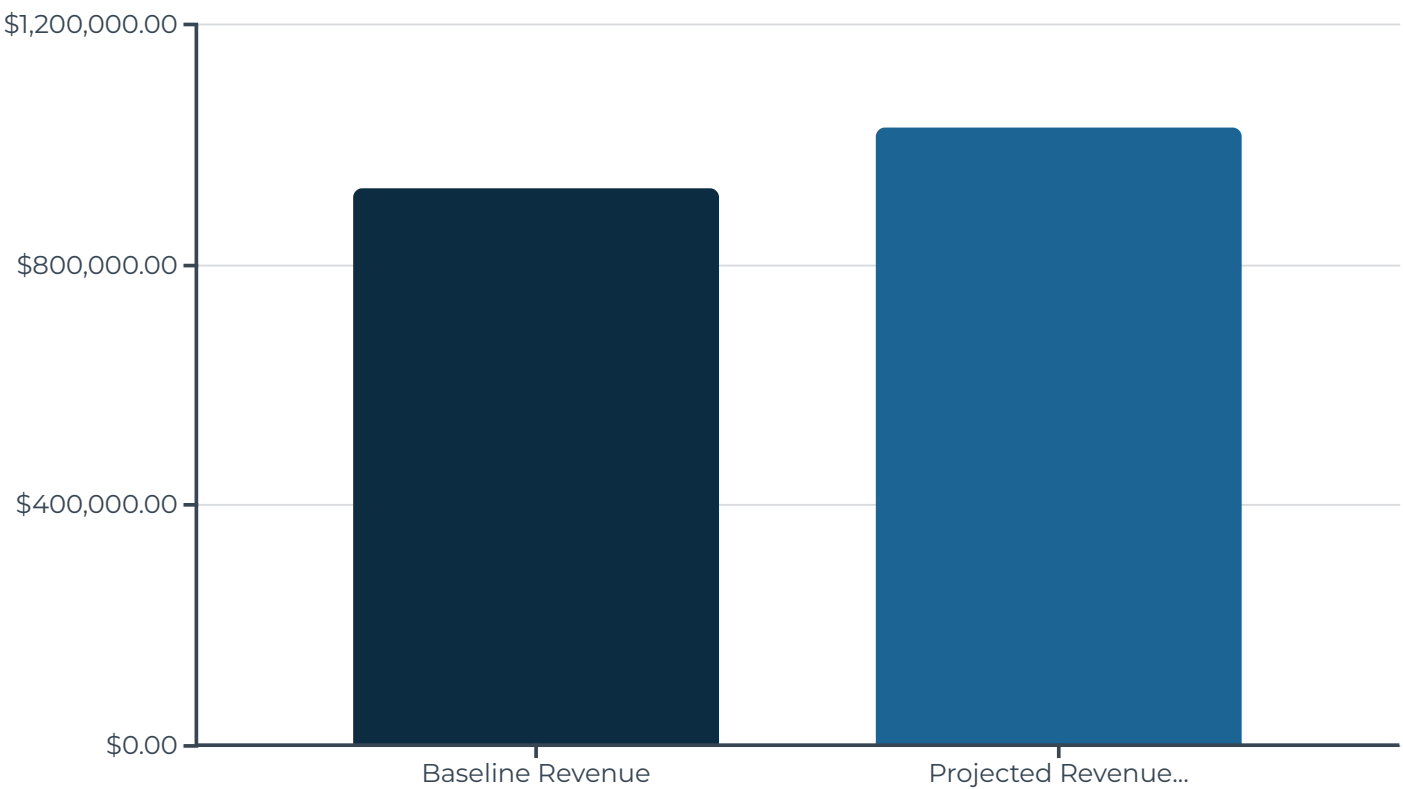
Premium Positioning

Showcase products and services that align with their high-income lifestyle, emphasizing value and exclusivity.

Competitor Differentiation

Analyze competitor offerings and develop superior value propositions that highlight our unique advantages.

ROI of Targeted Marketing (Visual Summary)



By implementing these focused strategies, we anticipate a substantial net gain and a high return on investment.

+\$76K

Net Gain

Additional annual revenue generated from targeted campaigns.

305%

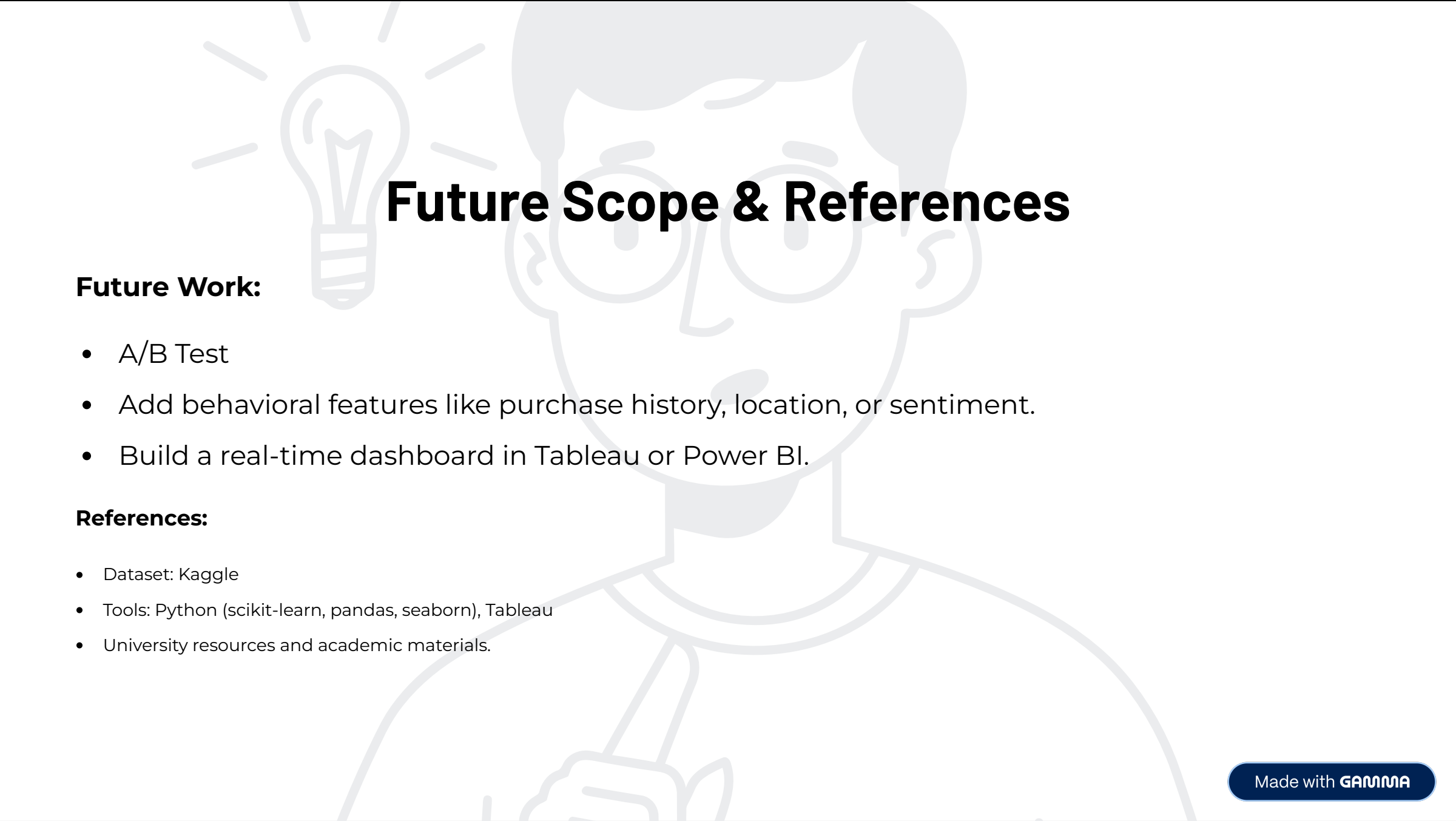
Return on Investment

Achieved by leveraging a campaign cost of \$25,000.



Expected Results & Conclusion

- The model successfully identified **key customer segments** using both **Hierarchical Clustering (5 groups)** and **K-Means (4 groups)**.
- The **K-Means** formed compact, centroid-based clusters for efficient segmentation, while **Hierarchical Clustering** revealed broader, distance-based linkages for deeper analysis.
- These insights enable **targeted marketing, optimized product placement**, and **customer retention** strategies.
- Provides a **scalable foundation** for future customer segmentation using Machine Learning.

A faint, stylized background illustration of a person with short dark hair and round glasses, looking upwards. To the left of the person's head is a lightbulb with several short lines radiating from it, indicating it is lit. The person is wearing a white shirt and has their right hand raised near their chin.

Future Scope & References

Future Work:

- A/B Test
- Add behavioral features like purchase history, location, or sentiment.
- Build a real-time dashboard in Tableau or Power BI.

References:

- Dataset: Kaggle
- Tools: Python (scikit-learn, pandas, seaborn), Tableau
- University resources and academic materials.